

Hundred Days of Coding in C Programing

Day – 18

Name – Arayana Khare

Batch – 15

Sap id – 590042432

Question

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, with void return type and with a simple function call (without call by value/reference). Each function should read the required input, process it, and display the result.

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

- 1. Check whether it is a Strong number.**
- 2. Check whether it is an Armstrong number.**
- 3. Check whether it is a Harshad number.**
- 4. Check whether it is a Duck number.**

Step – 1 Algorithm

- 1: Start the program.
- 2: Display the menu containing Strong Number, Armstrong Number, Harshad Number, Duck Number, and Exit.
- 3: Read the user's choice.
- 4: Use switch-case to call the appropriate function.
- 5: For Strong Number: extract each digit, find its factorial, add the factorials, and compare the sum with the original number.
- 6: For Armstrong Number: count the digits, raise each digit to the power of the number of digits, add the results, and compare the sum with the original number.
- 7: For Harshad Number: find the sum of the digits and check whether the original number is divisible by the digit sum.
- 8: For Duck Number: check whether the number contains the digit 0.
- 9: Repeat the menu until the user selects Exit.
- 10: Stop the program.

Step – 2 Test Case Table

<u>Test Case</u>	<u>Menu Choice</u>	<u>Input</u>	<u>Expected Output</u>
<u>1</u>	<u>1</u>	<u>145</u>	<u>145 is a Strong number</u>
<u>2</u>	<u>1</u>	<u>123</u>	<u>123 is not a Strong number</u>
<u>3</u>	<u>2</u>	<u>153</u>	<u>153 is an Armstrong number</u>
<u>4</u>	<u>2</u>	<u>123</u>	<u>123 is not an Armstrong number</u>
<u>5</u>	<u>3</u>	<u>18</u>	<u>18 is a Harshad number</u>
<u>6</u>	<u>3</u>	<u>19</u>	<u>19 is not a Harshad number</u>
<u>7</u>	<u>4</u>	<u>1023</u>	<u>1023 is a Duck number</u>
<u>8</u>	<u>4</u>	<u>1234</u>	<u>1234 is not a Duck number</u>

Step – 3 Write a Code in C Programing

```
#include <stdio.h>
```

```
/* Function to check Strong Number */
```

```
void strongNumber()
```

```
{
```

```
    int n, temp, digit, sum = 0;
```

```
    int fact, i;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &n);
```

```
    temp = n;
```

```
    while (temp > 0)
```

```
    {
```

```
        digit = temp % 10;
```

```
        fact = 1;
```

```
        for (i = 1; i <= digit; i++)
```

```
        {
```

```
            fact = fact * i;
```

```
        }
```

```
        sum = sum + fact;
```

```
        temp = temp / 10;
```

```
    }
```

```
    if (sum == n)
        printf("%d is a Strong number.\n", n);
    else
        printf("%d is not a Strong number.\n", n);
}
```

```
/* Function to check Armstrong Number */
```

```
void armstrongNumber()
```

```
{
```

```
    int n, temp, digit;
```

```
    int digits = 0, sum = 0;
```

```
    int power, i;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &n);
```

```
    temp = n;
```

```
    while (temp > 0)
```

```
    {
```

```
        digits++;
```

```
        temp = temp / 10;
```

```
    }
```

```
    temp = n;
```

```
    while (temp > 0)
```

```
{  
    digit = temp % 10;  
  
    power = 1;  
    for (i = 1; i <= digits; i++)  
    {  
        power = power * digit;  
    }  
  
    sum = sum + power;  
    temp = temp / 10;  
}  
  
if (sum == n)  
    printf("%d is an Armstrong number.\n", n);  
else  
    printf("%d is not an Armstrong number.\n", n);  
}
```

```
/* Function to check Harshad Number */
```

```
void harshadNumber()
```

```
{  
    int n, temp, digit, sum = 0;  
  
    printf("Enter a positive integer: ");  
    scanf("%d", &n);
```

```
temp = n;
```

```
while (temp > 0)
```

```
{
```

```
    digit = temp % 10;
```

```
    sum = sum + digit;
```

```
    temp = temp / 10;
```

```
}
```

```
if (n % sum == 0)
```

```
    printf("%d is a Harshad number.\n", n);
```

```
else
```

```
    printf("%d is not a Harshad number.\n", n);
```

```
}
```

```
/* Function to check Duck Number */
```

```
void duckNumber()
```

```
{
```

```
    int n, temp, digit;
```

```
    int found = 0;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &n);
```

```
    temp = n;
```

```
    while (temp > 0)
```

```

{
    digit = temp % 10;

    if (digit == 0)
    {
        found = 1;
        break;
    }

    temp = temp / 10;
}

if (found)
    printf("%d is a Duck number.\n", n);
else
    printf("%d is not a Duck number.\n", n);
}

/* Main Function */
int main()
{
    int choice;

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Check Strong Number\n");

```



```
printf("2. Check Armstrong Number\n");  
printf("3. Check Harshad Number\n");  
printf("4. Check Duck Number\n");  
printf("5. Exit\n");  
printf("=====\n");
```

```
printf("Enter your choice: ");  
scanf("%d", &choice);
```

```
switch (choice)  
{  
    case 1:  
        strongNumber();  
        break;  
  
    case 2:  
        armstrongNumber();  
        break;  
  
    case 3:  
        harshadNumber();  
        break;  
  
    case 4:  
        duckNumber();  
        break;  
  
    case 5:
```

```
        printf("Program terminated.\n");  
        break;  
  
    default:  
        printf("Invalid choice. Please try again.\n");  
    }  
  
} while (choice != 5);  
  
return 0;  
}
```

Step – 4 Compilation and Execution

Case: 1

```
148
149     switch (choice)
150     {
151         case 1:
152             strongNumber();
153             break;
154
155         case 2:
156             armstrongNumber();
157             break;
158
159         case 3:
160             harshadNumber();
161             break;
162
163         case 4:
164             duckNumber();
165             break;
166
167         case 5:
168             printf("Program terminated.\n");
169             break;
170
171         default:
172             printf("Invalid choice. Please try again.\n");
173     }
174
175     } while (choice != 5);
176
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 1
Enter a positive integer: 145
145 is a Strong number.
```

Case: 2

```
149     switch (choice)
150     {
151         case 1:
152             strongNumber();
153             break;
154
155         case 2:
156             armstrongNumber();
157             break;
158
159         case 3:
160             harshadNumber();
161             break;
162
163         case 4:
164             duckNumber();
165             break;
166
167         case 5:
168             printf("Program terminated.\n");
169             break;
170
171         default:
172             printf("Invalid choice. Please try again.\n");
173     }
174
175     } while (choice != 5);
176
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 1
Enter a positive integer: 123
123 is not a Strong number.
```

Case: 3

```
149         switch (choice)
150         {
151             case 1:
152                 strongNumber();
153                 break;
154
155             case 2:
156                 armstrongNumber();
157                 break;
158
159             case 3:
160                 harshadNumber();
161                 break;
162
163             case 4:
164                 duckNumber();
165                 break;
166
167             case 5:
168                 printf("Program terminated.\n");
169                 break;
170
171             default:
172                 printf("Invalid choice. Please try again.\n");
173         }
174
175     } while (choice != 5);
176
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 2
Enter a positive integer: 153
153 is an Armstrong number.
```

Case: 4

```
149         switch (choice)
150         {
151             case 1:
152                 strongNumber();
153                 break;
154
155             case 2:
156                 armstrongNumber();
157                 break;
158
159             case 3:
160                 harshadNumber();
161                 break;
162
163             case 4:
164                 duckNumber();
165                 break;
166
167             case 5:
168                 printf("Program terminated.\n");
169                 break;
170
171             default:
172                 printf("Invalid choice. Please try again.\n");
173         }
174     } while (choice != 5);
175
176
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 2
Enter a positive integer: 123
123 is not an Armstrong number.
```

Case: 5

```
149     switch (choice)
150     {
151         case 1:
152             strongNumber();
153             break;
154
155         case 2:
156             armstrongNumber();
157             break;
158
159         case 3:
160             harshadNumber();
161             break;
162
163         case 4:
164             duckNumber();
165             break;
166
167         case 5:
168             printf("Program terminated.\n");
169             break;
170
171         default:
172             printf("Invalid choice. Please try again.\n");
173     }
174
175     } while (choice != 5);
176
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 3
Enter a positive integer: 18
18 is a Harshad number.
```

Case: 6

```
149     switch (choice)
150     {
151         case 1:
152             strongNumber();
153             break;
154
155         case 2:
156             armstrongNumber();
157             break;
158
159         case 3:
160             harshadNumber();
161             break;
162
163         case 4:
164             duckNumber();
165             break;
166
167         case 5:
168             printf("Program terminated.\n");
169             break;
170
171         default:
172             printf("Invalid choice. Please try again.\n");
173     }
174
175     } while (choice != 5);
176
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 3
Enter a positive integer: 19
19 is not a Harshad number.
```


Case: 7

```
149     switch (choice)
150     {
151         case 1:
152             strongNumber();
153             break;
154
155         case 2:
156             armstrongNumber();
157             break;
158
159         case 3:
160             harshadNumber();
161             break;
162
163         case 4:
164             duckNumber();
165             break;
166
167         case 5:
168             printf("Program terminated.\n");
169             break;
170
171         default:
172             printf("Invalid choice. Please try again.\n");
173     }
174
175     } while (choice != 5);
176
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 4
Enter a positive integer: 1023
1023 is a Duck number.
```

Case: 8

```
151         case 1:
152             strongNumber();
153             break;
154
155         case 2:
156             armstrongNumber();
157             break;
158
159         case 3:
160             harshadNumber();
161             break;
162
163         case 4:
164             duckNumber();
165             break;
166
167         case 5:
168             printf("Program terminated.\n");
169             break;
170
171         default:
172             printf("Invalid choice. Please try again.\n");
173     }
174
175     } while (choice != 5);
176
177     return 0;
178 }
```

input

```
===== MENU =====
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
5. Exit
=====
Enter your choice: 4
Enter a positive integer: 1234
1234 is not a Duck number.
```

Step – 5 Conclusion

The menu-driven C program successfully performs Strong, Armstrong, Harshad, and Duck number checks using switch-case. Each operation is implemented using a separate void user-defined function with a simple function call and no arguments.